25900869 **Reg.No : .....**

Name : .....

MAHATMA GANDHI UNIVERSITY, KOTTAYAM

MGU-UGP (HONOURS) REGULAR /IMPROVEMENT/ REAPPEARANCE EXAMINATION NOVEMBER 2025

FIRST SEMESTER

Discipline Specific Core Course (DSC) - MG1DSCHPY100 - BASICS OF PHYSIOLOGY

(2024 ADMISION ONWARDS)

Duration: 2 Hours **Maximum Marks: 50**

Remember(K), Understand(U), Apply(A), Analyse(An), Evaluate(E), Create(C), Skill(S), Interest(I) and Appreciation(Ap)

Students should attempt at least one question from each course outcome to enhance their overall outcome attainability.

Part A

Multiple Choice Questions

Answer **All** Questions

Each Question carries **1** mark

1. Select the type of tissue responsible for transmitting signals throughout the body. [K] / [CO1]

   a). Connective tissue b). Epithelial tissue

   c). Muscular tissue d). Nervous tissue

2. The ratio between the percentage of RBC and haemoglobin is ------- [K] / [CO2]

   a). MCH b). CI

   c). MCV d). MCHC

3. The thoracic cage consists of ---------- [K] / [CO3]

   a). Ribs and sternum b). Ribs and vertebrae

   c). Ribs and scapulla d). Ribs and maxilla

4. Name the part of the brain is primarily responsible for regulating body temperature. [K] / [CO1]

   a). Cerebrum b). Cerebellum

c). Hypothalamus d). Medulla oblongata

5. The smallest component of ECF is ---------- [K] / [CO2]

a). Mesenchymal Fluid b). Intracellular fluid

c). Plasma d). Transcellular fluid

6. Osteocyte is located in a small cavity in the bone tissue called ---------- [U] / [CO3]

a). Bone marrow b). Matrix

c). Lacuna d). Canaliculi

[6x1=6]

Part B

Short answer Questions

Answer any **2** Questions

Each Question carries **2** marks

7. Describe the structure and function of adherens junctions. [K] / [CO1]

8. Illustrate the functions of RBC. [U] / [CO2]

9. Classify muscle depending on striations [U] / [CO3]

10. Define short bones [K] / [CO3]

[2x2=4]

Part C

Short Essay Questions

Answer any **4** Questions

Each Question carries **5** marks

11. Describe the structure and function of glandular epithelium. How are glands classified? [K] / [CO1]

12. Describe special types of passive transport [K] / [CO1]

13. Classify oedema. [U] / [CO2]

14. Explain the composition and formation of lymph [U] / [CO2]

15. Describe skeletal muscles with diagram [U] / [CO3]

16. Explain the bones of pectoral and pelvic girdle [U] / [CO3]

[4x5=20]

Part D

Long essay Questions

Answer any **2** Questions

Each Question carries **10** marks

17.	Explain cell signaling by describing its stages, types, and the role of signaling molecules.	[U] / [CO1]
18.	Explain structure and functions of lymphatic system	[K] / [CO2]
19.	Explain physiological and pathological variation in WBC count	[U] / [CO2]
20.	Explain structural classification of joints	[U] / [CO3]

[2x10=20]